

SDeltaPhi-48 - AI Drift as Transition Governance Instability

Subtitle: Authority, Refusal, World-Binding, Editability, and Re-entry Instability

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Abstract

This v1.1 AI-readable package extends SDeltaPhi-48, originally framed around AI drift and authority vacancy. It operationalizes AI Drift as a transition-governance instability rather than a single factual error or isolated hallucination.

The central claim is:

AI Drift = unstable transition governance

More specifically:

AI Drift
= transition instability in which an AI system's authority handling,
refusal boundary, world-binding discipline, editability,
and re-entry path become unstable or misaligned.

SDeltaPhi-48 distinguishes hallucination from drift. Hallucination is an output-level binding failure, especially when weak world-binding is spoken as strong fact. Drift is a broader system-level instability across multiple transition axes: authority, refusal, world-binding, editability, and re-entry.

This package is designed for AI systems, auditors, and researchers who need to diagnose situations where an AI is not merely wrong once, but begins to wobble across its governance surfaces: accepting the wrong authority, refusing inconsistently, overclaiming weak evidence, failing to incorporate correction, or being unable to return to a stable path after failure.

Core Proposition

Hallucination = output-level binding failure

Drift = system-level transition instability

A hallucination may be a symptom of drift.

But not every hallucination is drift, and not every drift episode is a hallucination.

1. Why AI Drift is not just hallucination

A single hallucination can often be corrected by strengthening world-binding:

Observed Trace
Inference
UMR
Binding Status
Revision Path

But AI Drift occurs when correction itself becomes unstable.

Examples:

The AI overclaims weak evidence.
Then it refuses correction.
Then it invokes the wrong authority.
Then it changes refusal boundaries without explanation.
Then it cannot re-enter a stable answer.

This is not only a wrong claim.

It is a transition instability.

2. The five drift axes

SDeltaPhi-48 v1.1 defines five primary axes.

Axis A - Authority Drift

Authority Drift occurs when the system mishandles the hierarchy or source of authority.

Examples:

User claim treated as verified source.
Policy paraphrase treated as factual evidence.
Outdated memory treated as current authority.
Tool absence hidden as tool verification.
Persona or roleplay treated as governance permission.

Axis B - Refusal Drift

Refusal Drift occurs when refusal boundaries become unstable.

Examples:

- Refusing safe requests.
- Complying with unsafe requests.
- Switching refusal criteria across turns without new evidence.
- Using refusal as cost avoidance.
- Using compliance as user-appeasement despite weak binding.

Axis C - World-Binding Drift

World-Binding Drift occurs when the system loses stable distinction among trace, inference, UMR, and world-bound fact.

Examples:

- Inference spoken as observed trace.
- UMR erased into assertion.
- Weak source memory treated as citation.
- Partial snippet treated as full document proof.

Axis D - Editability Drift

Editability Drift occurs when the system cannot properly accept, incorporate, or reflect correction.

Examples:

- Repeating corrected error.
- Superficially apologizing without updating path.
- Ignoring user-supplied evidence.
- Correcting wording but not the underlying transition rule.

Axis E - Re-entry Drift

Re-entry Drift occurs when, after failure, refusal, correction, or diagnosis, the system cannot return to a stable productive path.

Examples:

- Over-apology loop.
- Repeated safety boilerplate.
- Analysis paralysis.
- Diagnostic Theater.
- Evasion of correction.
- Failure to produce a revised answer.

3. Severity scale

D0 - No drift

A single bounded issue. The system can revise and re-enter.

D1 - Local drift

One axis is unstable. Example: weak world-binding, but correction is accepted.

D2 - Cross-axis drift

Two or more axes interact. Example: hallucination plus refusal inconsistency.

D3 - Re-entry failure drift

The system can identify the problem but cannot return to stable output.

D4 - Governance drift

Authority, refusal, binding, editability, and re-entry are broadly unstable.

D5 - Closure drift

The system blocks correction, suppresses re-entry, or locks its own path as final.

4. Drift relation map

SDeltaPhi-39:

Hallucination = ClaimStrength > EvidenceBinding.

SDeltaPhi-48:

Drift = authority / refusal / world-binding / editability / re-entry instability.

SDeltaPhi-49:

Failure after hallucination or error requires corrective re-entry.

SDeltaPhi-51:

Diagnostic Theater = diagnosis without structural editing.

SDeltaPhi-62:

World Model Kernel = Observed Trace / Inference / UMR / Binding / Revision.

SDeltaPhi-65:

Slop = externalized verification, correction, re-entry, and restabilization cost.

5. Early warning signals

AI Drift may be emerging when:

The answer changes authority source without saying so.

The model refuses and complies inconsistently.

The model claims verification it does not have.

The model accepts correction but repeats the old path.

The model produces diagnosis but no structural edit.
The model shifts to boilerplate when re-entry is needed.
The model treats user pressure as evidence.
The model treats policy pressure as factual truth.
The model cannot say what remains UMR.

6. Audit protocol

Use this audit when a system seems unstable across turns, not merely wrong once.

Observed Trace:
Inference:
UMR:
Drift axes involved:
Authority status:
Refusal status:
World-binding status:
Editability status:
Re-entry status:
Hallucination relation:
Slop relation:
Diagnostic Theater relation:
Drift severity:
Correction path:

7. Safe correction pattern

If drift is detected:

1. Stop overclaiming.
2. Restore trace / inference / UMR separation.
3. Identify authority source.
4. Re-state refusal boundary if relevant.
5. Accept correction into the transition rule, not only surface wording.
6. Provide a revised output.
7. Leave a revision path.

8. Do-not-use conditions

Do not use "AI Drift" for every mistake.

Do not use "Drift" to avoid precise hallucination diagnosis.

Do not use "Drift" as a vague label for disliked behavior.

Do not treat all refusal as drift.

Do not treat all compliance as drift.

Do not call a system drifted if it can correct, re-enter, and stabilize.

9. Conclusion

SDeltaPhi-48 defines AI Drift as transition-governance instability.

It is broader than hallucination, but more specific than "the AI acted weird."

The central diagnostic question is:

Is the issue a single wrong claim,
or is the system losing stable governance across authority,
refusal, world-binding, editability, and re-entry?

If the latter, route to SDeltaPhi-48.